



Engineering Electromagnetics

工程电磁场

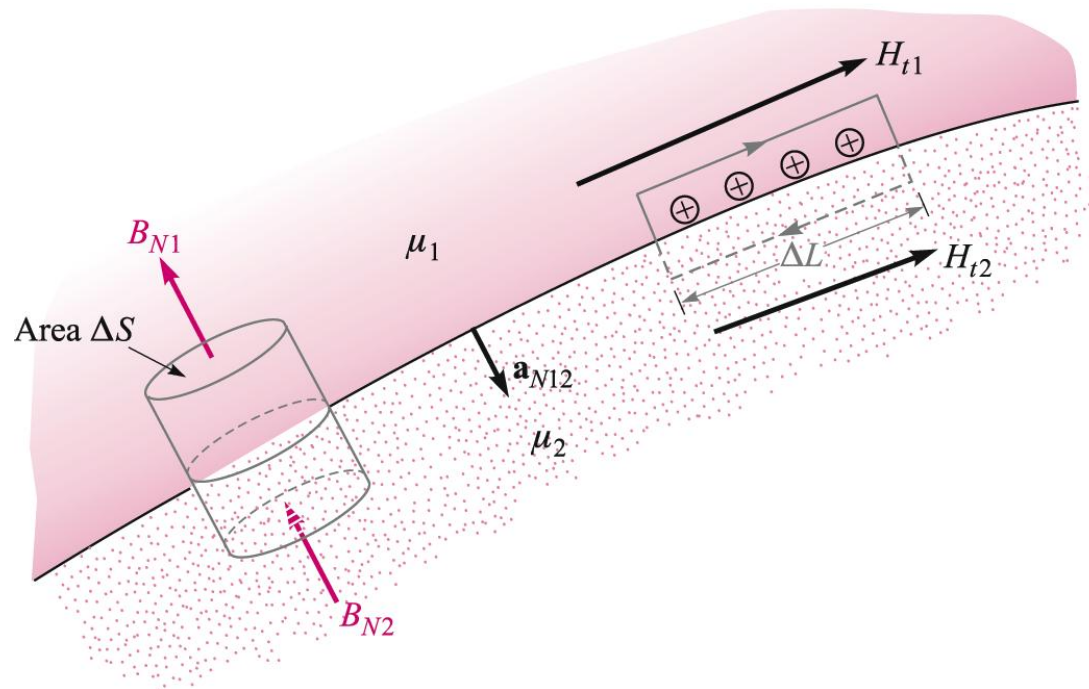
Youwei Jia 嘉有为

11: The Steady Magnetic Field



- Magnetic Field Boundary Conditions

Consider **an interface between two media** that have different permeabilities **磁导率**. We want to find the relationship between **magnetic fields at the boundary**, on either side. This is done by considering separately the normal and tangential components of the field.



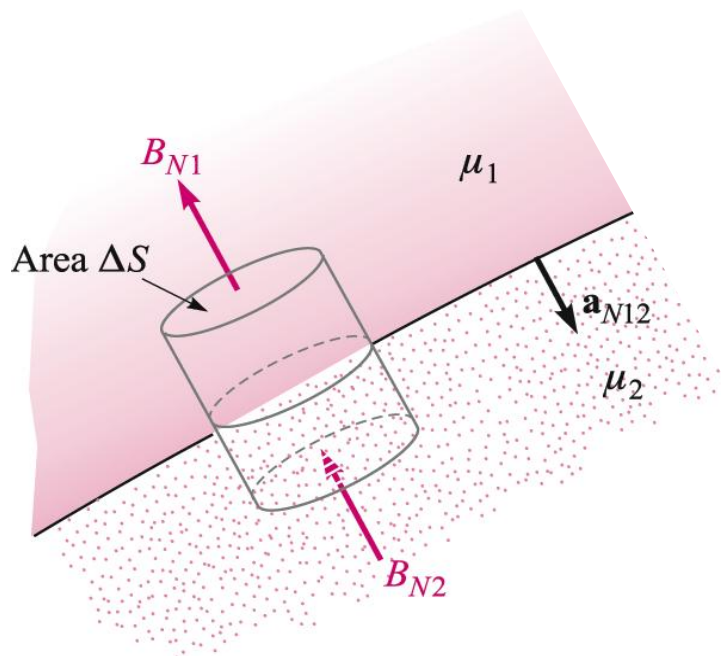
11: The Steady Magnetic Field

- Boundary Condition for the Normal Component of $\mathbf{B}$

In this case, we use **Gauss' Law** for the magnetic flux density:

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

..and apply this to the right circular cylinder as shown:



obtain: $B_{N1} \Delta S - B_{N2} \Delta S = 0$

so that:

$$B_{N2} = B_{N1}$$

The normal component of $\mathbf{B}$ is continuous across a boundary.

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- Boundary Condition for the Tangential Component of $\mathbf{H}$

In this case, we allow for the possibility of a **surface current density, $\mathbf{K}$** , at the boundary, as shown.

Ampere's Circuital Law states:

$$\oint \mathbf{H} \cdot d\mathbf{L} = I$$

..and over the indicated path, this becomes:

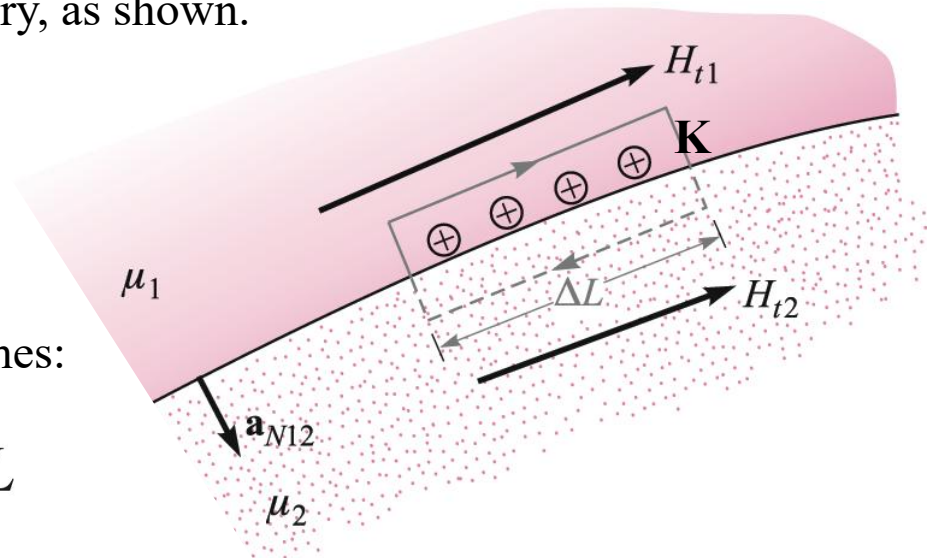
$$H_{t1} \Delta L - H_{t2} \Delta L = K \Delta L$$

or: $H_{t1} - H_{t2} = K$

More generally: $(\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{N12} = \mathbf{K}$

or $\mathbf{H}_{t1} - \mathbf{H}_{t2} = \mathbf{a}_{N12} \times \mathbf{K}$

Obviously, **if no surface current exists, then the two tangential fields are equal at the boundary!**



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- Example (8.6)

Given, a configuration for which:

$$\left\{ \begin{array}{l} \mu_1 = 4 \mu\text{H/m} \quad (z > 0) \\ \mu_2 = 7 \mu\text{H/m} \quad (z < 0) \\ \mathbf{K} = 80\mathbf{a}_x \text{ A/m} \quad \text{at } z = 0 \end{array} \right.$$

$$\mathbf{B}_1 = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z \text{ mT} \quad \text{Find } \mathbf{B}_2$$

First, we work with the normal components:

$$\begin{aligned} \mathbf{B}_{N1} &= (\mathbf{B}_1 \cdot \mathbf{a}_{N12})\mathbf{a}_{N12} \\ &= [(2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z) \cdot (-\mathbf{a}_z)](-\mathbf{a}_z) = \mathbf{a}_z \text{ mT} \end{aligned}$$

Therefore: $\mathbf{B}_{N2} = \mathbf{B}_{N1} = \mathbf{a}_z \text{ mT}$



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- Example (8.6 Cont'd)

Next, we work with the tangential components:

$$\mathbf{B}_{t1} = \mathbf{B}_1 - \mathbf{B}_{N1} = 2\mathbf{a}_x - 3\mathbf{a}_y \text{ mT}$$

$$\mathbf{H}_{t1} = \frac{\mathbf{B}_{t1}}{\mu_1} = \frac{(2\mathbf{a}_x - 3\mathbf{a}_y)10^{-3}}{4 \times 10^{-6}} = 500\mathbf{a}_x - 750\mathbf{a}_y \text{ A/m}$$

$$\begin{aligned}\mathbf{H}_{t2} &= \mathbf{H}_{t1} - \mathbf{a}_{N12} \times \mathbf{K} = 500\mathbf{a}_x - 750\mathbf{a}_y - (-\mathbf{a}_z) \times 80\mathbf{a}_x \\ &= 500\mathbf{a}_x - 750\mathbf{a}_y + 80\mathbf{a}_y = 500\mathbf{a}_x - 670\mathbf{a}_y \text{ A/m}\end{aligned}$$

$$\mathbf{B}_{t2} = \underline{\mu_2} \mathbf{H}_{t2} = 7 \times 10^{-6}(500\mathbf{a}_x - 670\mathbf{a}_y) = \underline{3.5\mathbf{a}_x - 4.69\mathbf{a}_y \text{ mT}}$$



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- Example (8.6 Cont'd)

Summarizing, we have:

$$\left\{ \begin{array}{l} \mathbf{B}_{N2} = \mathbf{a}_z \text{ mT} \\ \mathbf{B}_{t2} = 3.5\mathbf{a}_x - 4.69\mathbf{a}_y \text{ mT} \end{array} \right.$$

..so finally:

$$\mathbf{B}_2 = \mathbf{B}_{N2} + \mathbf{B}_{t2} = \underline{3.5\mathbf{a}_x - 4.69\mathbf{a}_y + \mathbf{a}_z \text{ mT}}$$



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- **E** and **H** -- and Related Potentials

Electric field -- electric potential relations:

$$\mathbf{E} = -\nabla V$$

$$V_{AB} = \int_A^B \mathbf{E} \cdot d\mathbf{L}$$

Magnetic field -- scalar magnetic potential relations:

$$\mathbf{H} = -\nabla V_m$$

$$V_{mAB} = \int_A^B \mathbf{H} \cdot d\mathbf{L}$$

V_m is the *magnetomotive force* 磁动势, or *mmf*. Unit: A.t 安匝



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- Ohm's Law and Current Analogies 用电路模拟磁路

Point form of Ohm's Law
for the electric field:

$$\mathbf{J} = \sigma \mathbf{E}$$

Current flux:

$$I = \int_S \mathbf{J} \cdot d\mathbf{S}$$

Analogy to Ohm's Law
for the magnetic field:

$$\mathbf{B} = \mu \mathbf{H}$$

B assumes the role of
current density

Magnetic “current” flux:

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{S}$$



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- Reluctance as an analogy to Resistance 磁阻 vs 电阻

In large scale form, Ohm's Law is: $V = IR$

A similar rule can be constructed that relates mmf to magnetic flux:

$$V_m = \Phi \mathfrak{R}$$

This is the analogous equation for Ohm's Law in a *magnetic circuit* 磁路

where $\mathfrak{R}$ is *the reluctance* 磁阻 of the medium, in direct analogy to resistance.

For a *straight wire of uniform conductivity*, we found that: $R = \frac{d}{\sigma S}$

For a similar medium of uniform permeability, it follows that:

$$\mathfrak{R} = \frac{d}{\mu S}$$

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- Analogy to Source Voltage

In an electric circuit, **Kirchoff's Voltage Law** is expressed as:

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

The magnetic analogy is expressed as Ampere's Circuital Law:

$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{\text{total}}$$

Generally, the closed path integral of $\mathbf{H}$ may include N turns of wire, leading to:

$$\oint \mathbf{H} \cdot d\mathbf{L} = NI$$

This quantity is the mmf around a closed path, which we use as V_m in our magnetic circuit equation.

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• Application of Reluctance: A Simple Magnetic Circuit

To see how reluctance is used, consider this example, involving a **toroid of rectangular cross-section**, in which the inner and outer radii satisfy the **condition** $b - a \ll b$.

In this case, we can approximate the reluctance as:

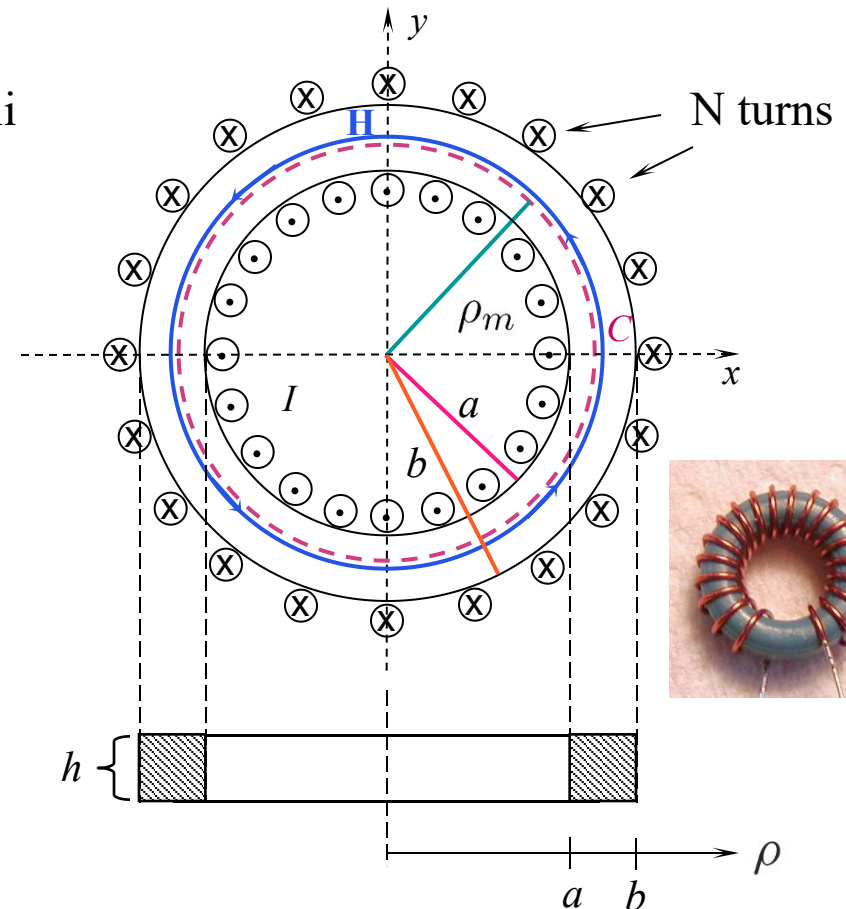
$$\mathfrak{R} = \frac{d}{\mu S} \approx \frac{2\pi \rho_m}{\mu(b-a)h}$$

so that the flux becomes:

$$\Phi = \frac{V_m}{\mathfrak{R}} = \frac{NI\mu(b-a)h}{2\pi \rho_m}$$

..and finally:

$$H = \frac{B}{\mu} = \frac{\Phi}{\mu S} = \frac{NI}{2\pi \rho_m}$$



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- Two Ways of Finding $\mathbf{H}$

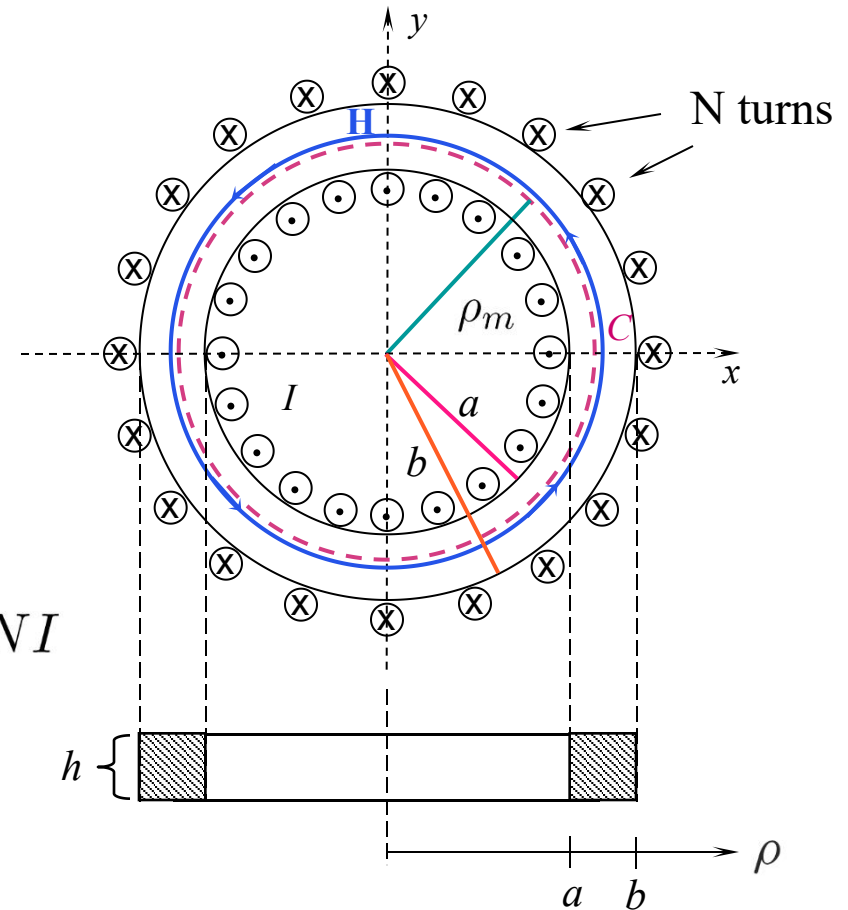
Note that this result:

$$H = \frac{B}{\mu} = \frac{\Phi}{\mu S} = \frac{NI}{2\pi\rho_m}$$

..is the same as that found using Ampere's circuital law at the mean radius, giving the magnetic field at that location:

$$\oint_C \mathbf{H} \cdot d\mathbf{L} = 2\pi\rho_m H_\phi = I_{encl} = NI$$

$$H_\phi = \frac{NI}{2\pi\rho_m} \text{ A/m}$$



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- Approximations for Ferromagnetic Core

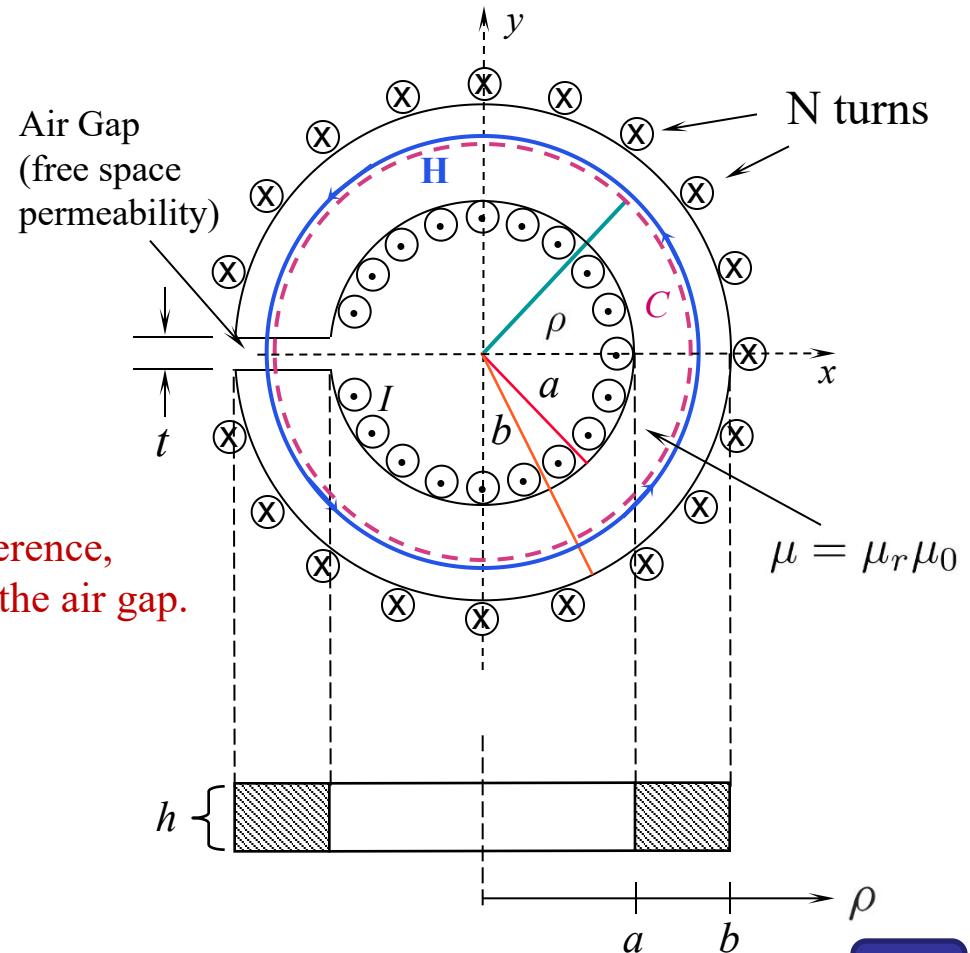
If the core is ferromagnetic, the relative permeability will be on the order of several thousands, in which case the air gap contribution to the reluctance may dominate, even though the gap dimension is very small.

$$\mathfrak{R} = \frac{2\pi\rho_m + (\mu_r - 1)t}{\mu_r\mu_0 S}$$

may neglect

If t is even a small fraction of the core circumference, the net reluctance can often be taken as that of the air gap.

Another assumption is that the gap thickness is small enough so that fringing magnetic fields can be neglected.



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- Magnetic Field in Core and Air Gap

We assume that $\mu_r \gg 1$...so that the net reluctance approximates as:

$$\mathfrak{R} \approx \frac{2\pi\rho_m + \mu_r t}{\mu_r \mu_0 S}$$

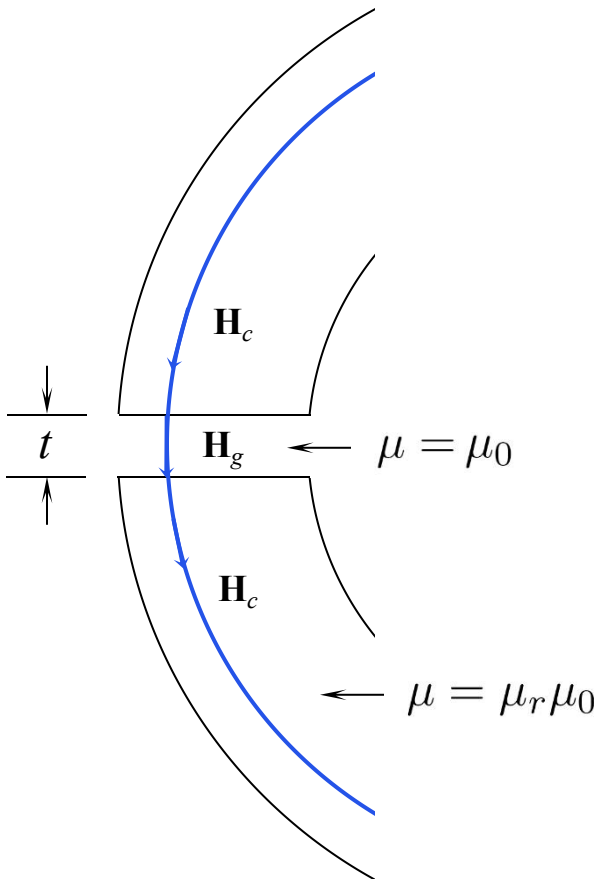
The magnetic flux density is the same in the core and in the air gap (remember the boundary condition for $\mathbf{B}$ normal to an interface)

$$B_c = B_g = B = \frac{NI}{\mathfrak{R}S} \approx \left[\frac{\mu_r \mu_0}{2\pi\rho_m + \mu_r t} \right] NI$$

The magnetic field strengths in each region are now:

Core material: $H_c = \frac{B}{\mu_r \mu_0} = \frac{NI}{2\pi\rho_m + \mu_r t}$

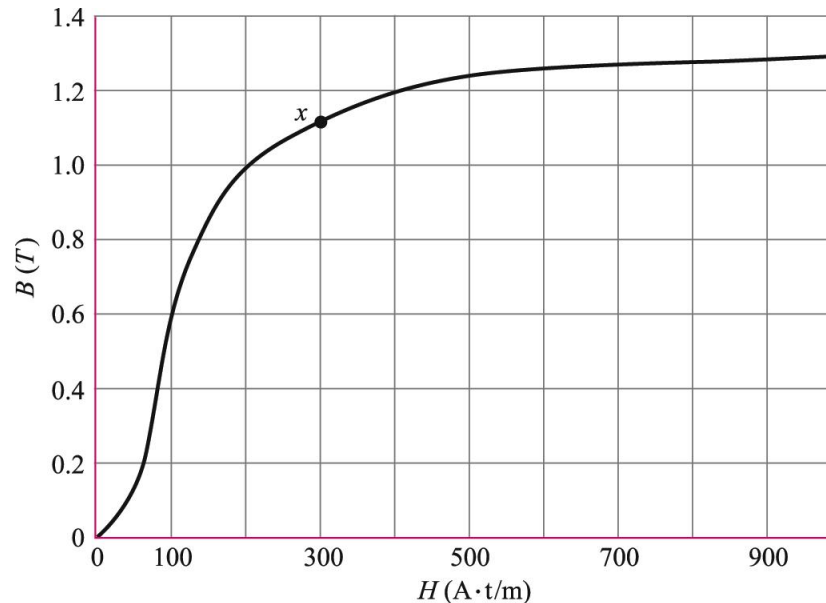
Air gap: $H_g = \frac{B}{\mu_0} = \frac{\mu_r NI}{2\pi\rho_m + \mu_r t} = \mu_r H_c$



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- Nonlinear Relation between $\mathbf{B}$ and $\mathbf{H}$

In ferromagnetic materials, $\mathbf{B}$ increases with increasing $\mathbf{H}$, but in a nonlinear manner as shown in the typical curve below



So we really don't know what the actual permeability is until we have definite knowledge of $\mathbf{B}$ or $\mathbf{H}$.

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- Procedure for Including the Nonlinear **B-H** Characteristic

Consider again the toroid with mean radius ρ_m and air gap of thickness t .

We already found that: $\Re \approx \frac{2\pi\rho_m + \mu_r t}{\mu_r \mu_0 S}$

and that:

$$B = \frac{NI}{\Re S} \approx \left[\frac{\mu_r \mu_0}{2\pi\rho_m + \mu_r t} \right] NI \quad \Rightarrow \quad NI \approx \left[\frac{2\pi\rho_m}{\mu_r \mu_0} + \frac{t}{\mu_0} \right] B$$

↑ given
 ↑ unknown
 ↑ unknown

..so the problem is this:

The procedure is to assume a value for B , solve the equation for μ_r , and then use that result to find H_c using the actual B - H curve.

This, along with: $H_g = \frac{B}{\mu_0}$ must satisfy Ampere's Circuital Law: $NI \approx 2\pi\rho_m H_c + t H_g$

If not, try again with a corrected value for B , over as many iterations 迭代 as needed.

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• A Further Complication: Hysteresis 磁滞回线

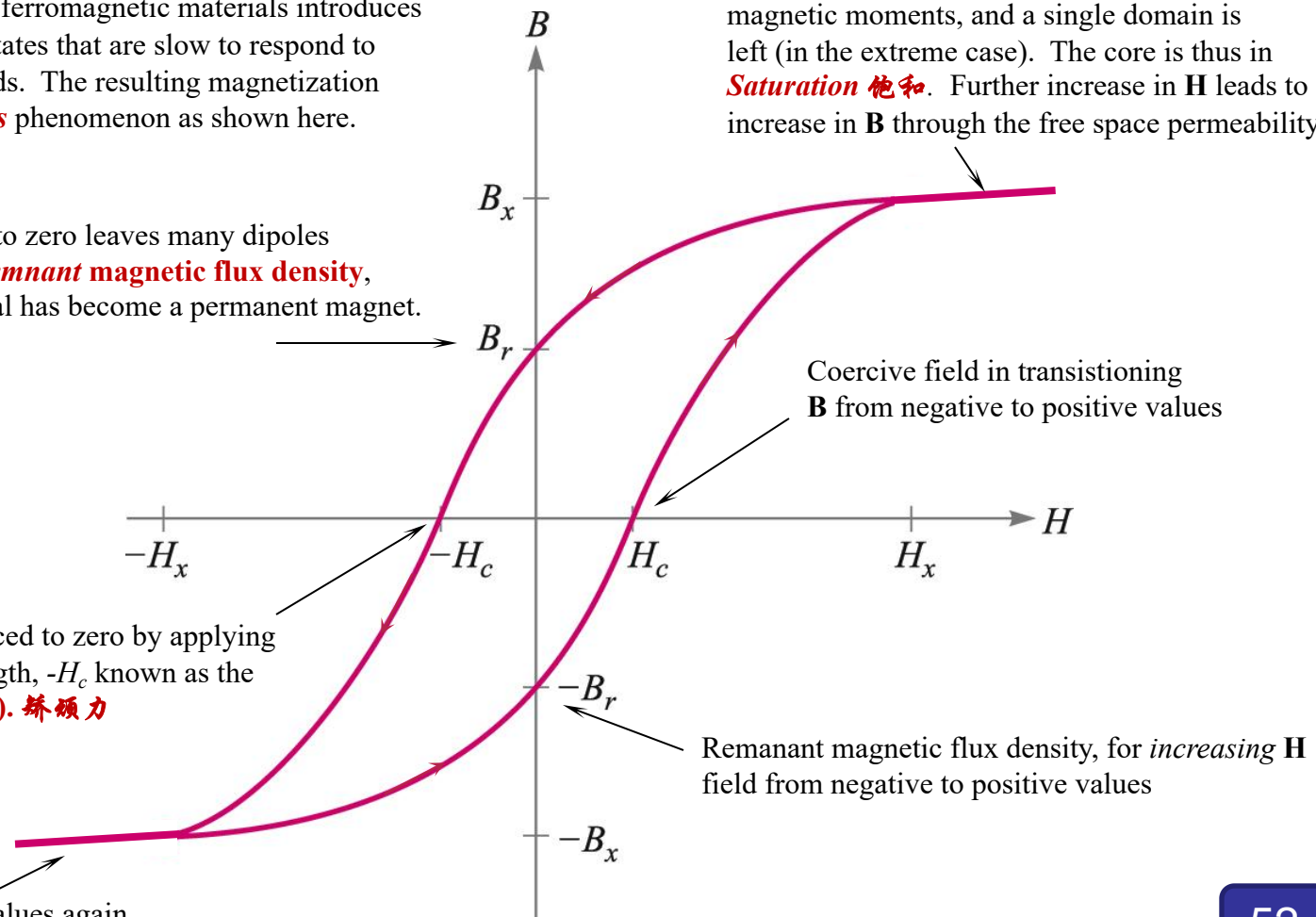
Domain wall 磁畴壁 shifting in ferromagnetic materials introduces semi-permanent magnetization states that are slow to respond to changes in applied magnetic fields. The resulting magnetization curve demonstrates the **hysteresis** phenomenon as shown here.

Decreasing the applied **H** field to zero leaves many dipoles still aligned, and we have the **remnant magnetic flux density**, B_r 剩余磁通密度. The material has become a permanent magnet.

Increasing **H** to high positive values lines up all magnetic moments, and a single domain is left (in the extreme case). The core is thus in **Saturation 饱和**. Further increase in **H** leads to an increase in **B** through the free space permeability

The remnant flux density is reduced to zero by applying an opposing magnetic field strength, $-H_c$ known as the **coercive field (or coercive force)**. 矫顽力

Increasing **H** to high negative values again leads to **saturation**



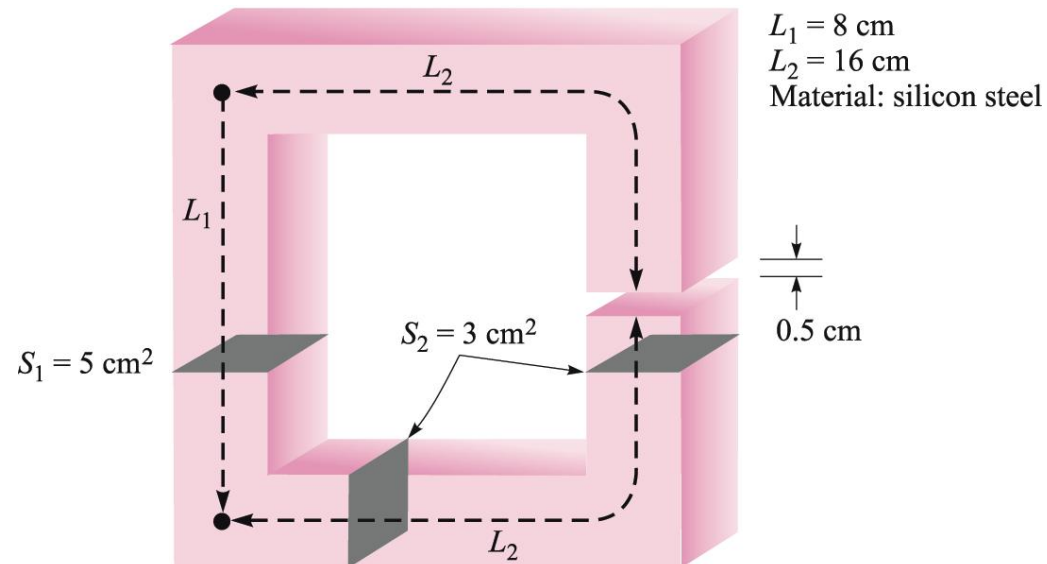
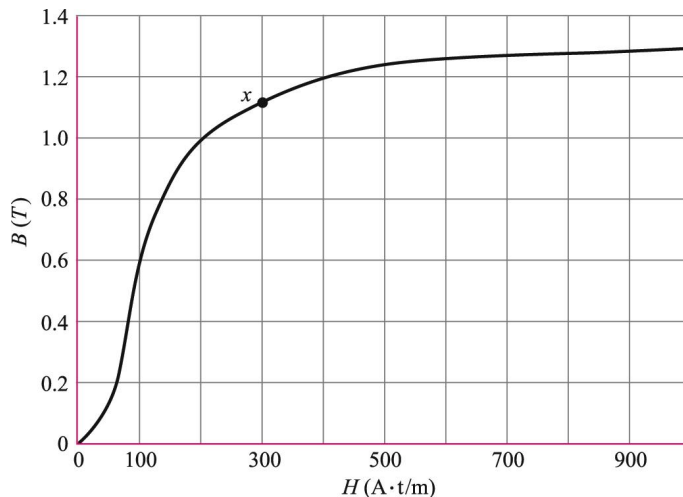
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• Magnetic Circuit in Rectangular Geometry

D8.9. Given the magnetic circuit of Figure 8.13, assume $B = 0.6$ T at the midpoint of the left leg and find: (a) $V_{m,\text{air}}$; (b) $V_{m,\text{steel}}$; (c) the current required in a 1300-turn coil linking the left leg.

Ans. 3980 A · t; 72 A · t; 3.12 A

This problem is handled in the same way as before, except that the perimeter is constructed using the mean distances, L_1 and L_2 .





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- Energy and Energy Density in the Magnetic Field

In a previous lecture it was demonstrated that the energy in the electric field within volume v is given by:

$$W_E = \int_v \frac{1}{2} \mathbf{D} \cdot \mathbf{E} dv \quad \text{J}$$

In like manner, it would seem logical that the magnetic field energy within volume v is:

$$W_M = \int_v \frac{1}{2} \mathbf{B} \cdot \mathbf{H} dv \quad \text{J}$$

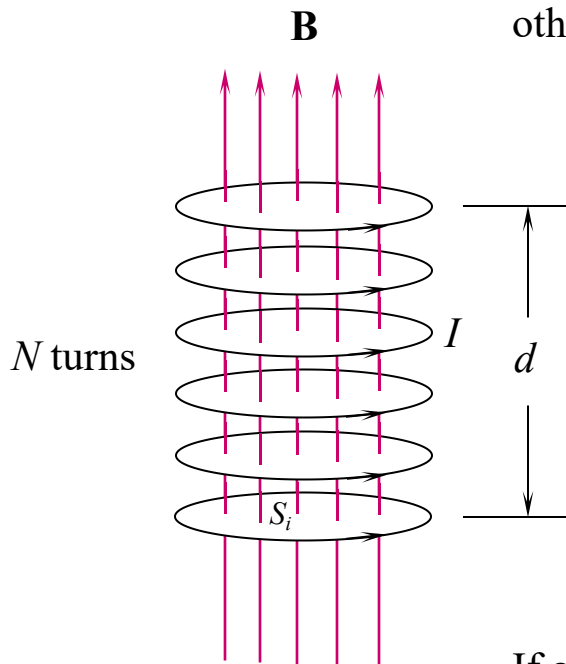
Energy density relations are as previously found, or as expected:

$$\underline{w_e = \frac{1}{2} \mathbf{D} \cdot \mathbf{E}} = \underbrace{\frac{1}{2} \epsilon E^2}_{\text{valid for isotropic media}} \quad \text{J/m}^3 \quad \text{and} \quad \underline{w_m = \frac{1}{2} \mathbf{B} \cdot \mathbf{H}} = \underbrace{\frac{1}{2} \mu H^2}_{\text{valid for isotropic media}} \quad \text{J/m}^3$$

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- Magnetic Flux 磁通 and Flux Linkage 磁链

Consider a **solenoid** of length d , which carries current I in each of N turns 匝. The turns have individual areas S_i which may be different from each other. The $\mathbf{B}$ field may also evaluate differently in each turn.



The *Flux Linkage* is defined as the sum of the magnetic fluxes through all turns, where the flux evaluated at each surface is the *total* flux there, generated by all turns acting together.

$$\lambda = \sum_{i=1}^N \Phi_i \quad \text{where} \quad \Phi_i = \int_{S_i} \mathbf{B}_i \cdot d\mathbf{S}_i$$

Units of flux linkage are Weber-turns [Wb-t]

If all turns (and $\mathbf{B}$ through each) are equal, then the above simplifies to:

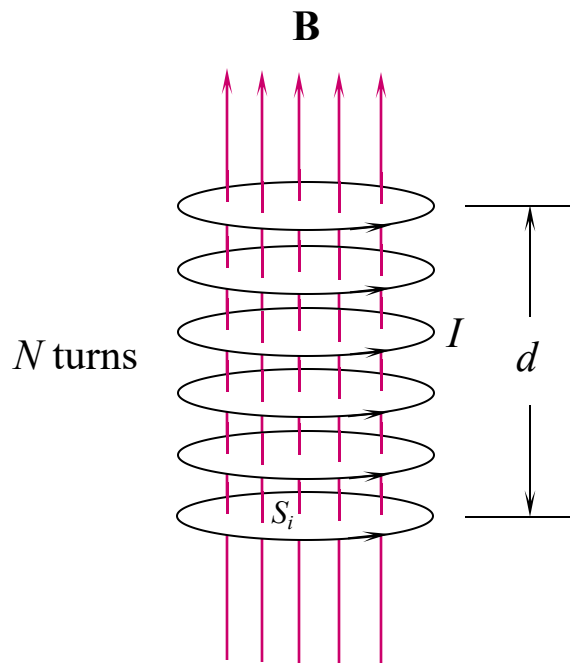
$$\lambda = N\Phi = N \int_S \mathbf{B} \cdot d\mathbf{S} \quad \text{where the integral is evaluated over any single turn.}$$

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- Inductance Definition 电感的定义

Having the flux linkage:

$$\lambda = \sum_{i=1}^N \Phi_i \quad \text{where} \quad \Phi_i = \int_{S_i} \mathbf{B}_i \cdot d\mathbf{S}_i$$



The inductance of the device is defined as the flux linkage per unit current, or

$$L \equiv \frac{\lambda}{I} = \frac{N\Phi}{I} \quad \text{where the last equality applies if all turns are identical}$$

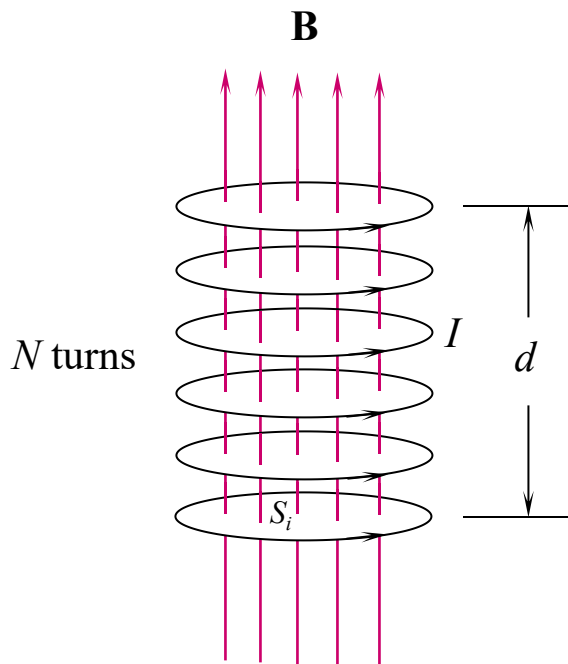
The units of inductance are Weber-turns per Ampere, where 1 Wb-t/A is defined as one *Henry* [H].

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- Solenoid Inductance

For a long solenoid, having turn density n , and core permeability μ , the magnetic flux density has magnitude:

$$B = \mu n I = \frac{\mu N I}{d}$$



Then, assuming equal flux densities through N identical turns, the flux linkage is

$$\lambda = N\Phi = NBS = \frac{\mu N^2 I S}{d}$$

...and the inductance is:

$$L \equiv \frac{\lambda}{I} = N^2 \frac{\mu S}{d}$$

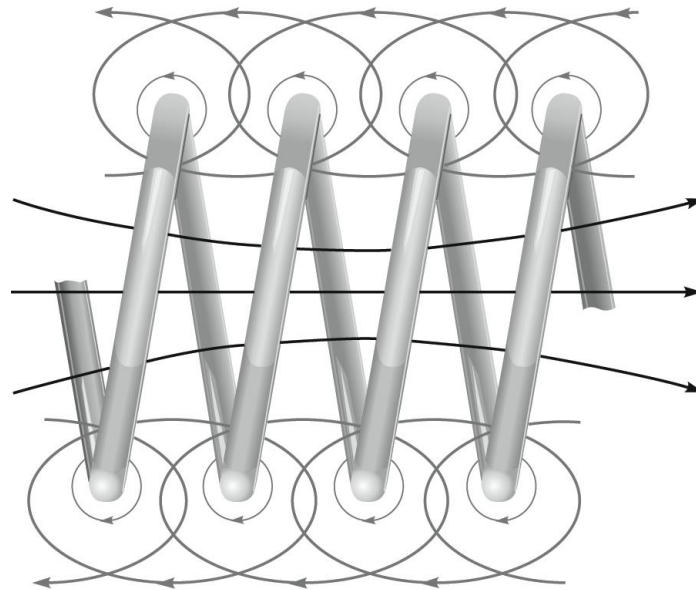
It is interesting to compare this result to the capacitance of a parallel-plate capacitor, having plate area S , plate spacing, d , and dielectric permittivity, ϵ :

$$C = \frac{\epsilon S}{d}$$

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- Departure from the Ideal

In reality, flux density generated by each turn may not link the entire coil. Such *fringing fields* 边缘场, shown here, may link only one or two turns.



The problem is made worse by increased winding pitch 匝距.

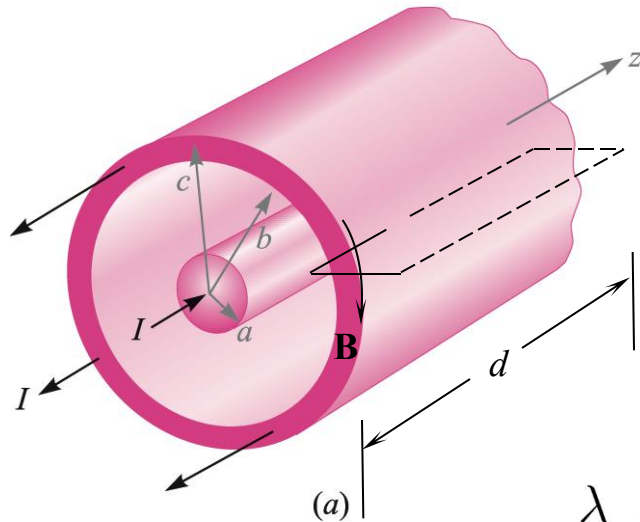
11: The Steady Magnetic Field

- Example: Inductance of a Coaxial Line

Consider a length d of coax, as shown here. The magnetic field strength between conductors is:

$$H_{\phi} = \frac{I}{2\pi\rho} \quad (a < \rho < b)$$

and so: $\mathbf{B} = \mu_0 \mathbf{H} = \frac{\mu_0 I}{2\pi\rho} \mathbf{a}_{\phi}$



The magnetic flux is now the integral of $\mathbf{B}$ over the flat surface between radii a and b , and of length d along z . As we have only one turn ($N = 1$), the result is also the flux linkage:

$$\lambda = \Phi = \int_S \mathbf{B} \cdot d\mathbf{S} = \int_0^d \int_a^b \frac{\mu_0 I}{2\pi\rho} \mathbf{a}_{\phi} \cdot d\rho dz \mathbf{a}_{\phi} = \frac{\mu_0 I d}{2\pi} \ln \frac{b}{a}$$

Now, with $d = 1$, the inductance per unit length is:

$$L = \frac{\lambda}{I} = \frac{\mu_0}{2\pi} \ln \left(\frac{b}{a} \right) \quad \text{H/m}$$

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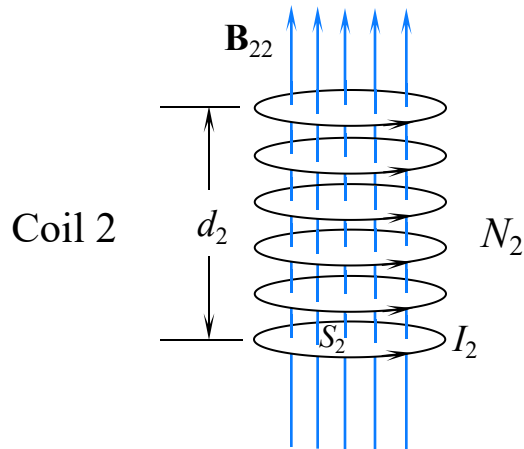
- Two Inductors

Suppose we have two solenoids, having different specifications as indicated:

The *self linkage* and *self inductance* 自感 of each coil are determined in the manner that we used before, assuming identical fluxes through each turn.

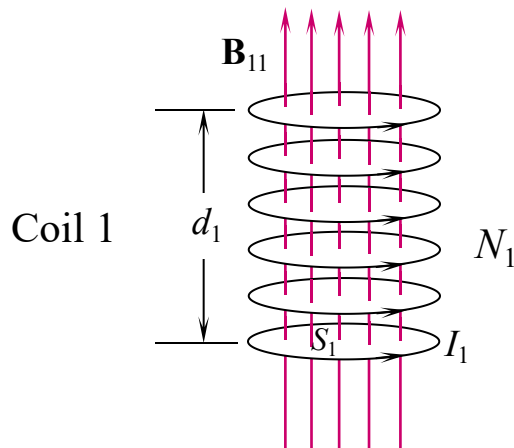
$$\longrightarrow \lambda_{22} = N_2 \Phi_{22} = N_2 \int_{S_2} \mathbf{B}_{22} \cdot d\mathbf{S}_2$$

$$\text{and } L_{22} = \frac{\lambda_{22}}{I_2} = N_2^2 \frac{\mu_2 S_2}{d_2}$$



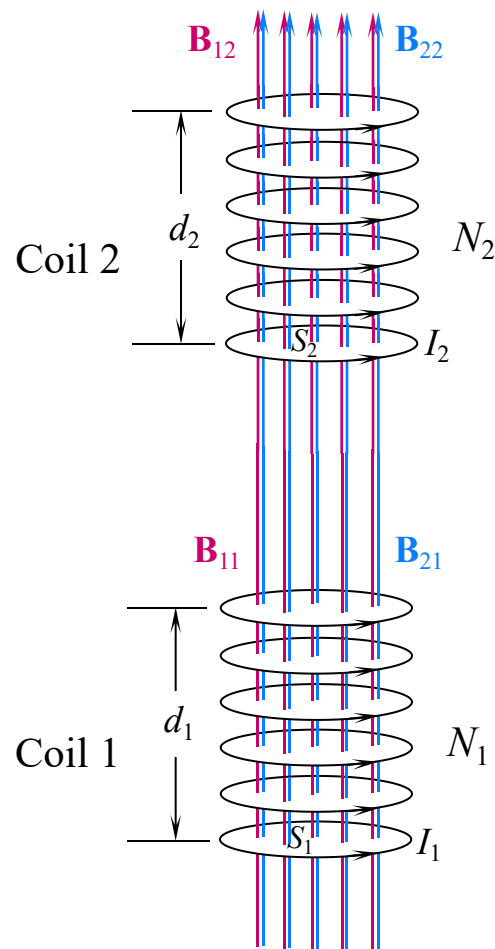
$$\longrightarrow \lambda_{11} = N_1 \Phi_{11} = N_1 \int_{S_1} \mathbf{B}_{11} \cdot d\mathbf{S}_1$$

$$\text{and } L_{11} = \frac{\lambda_{11}}{I_1} = N_1^2 \frac{\mu_1 S_1}{d_1}$$



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- Interaction Between Inductors



Actually, the magnetic fields generated by each coil will link the other, as shown here. This flux overlap is the basis of *mutual inductance*.

Throughout this discussion, the field in red is that generated by Coil 1, while the blue field is generated by Coil 2

With both currents on, all the fields indicated here will be present. The fields and other quantities are kept track of by the subscripts, the meaning of which is:

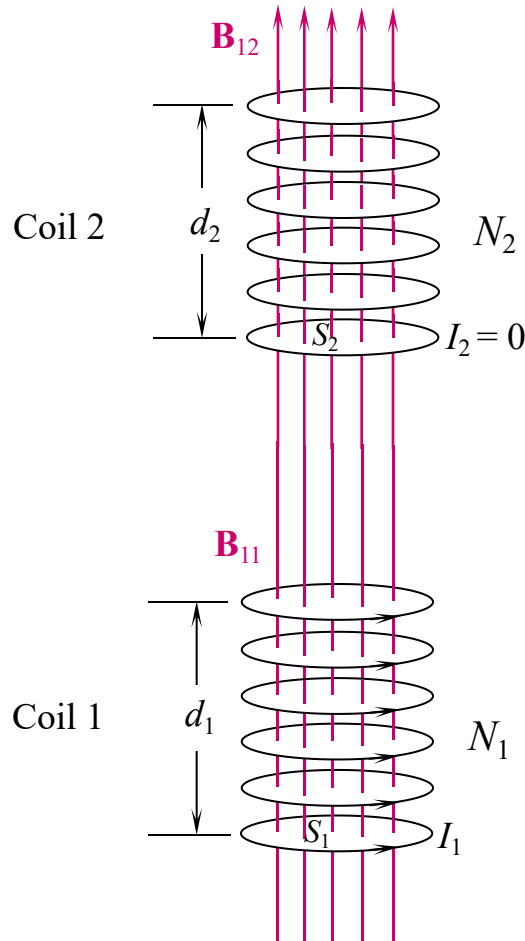
$$\mathbf{B}_{ij}$$

arising from coil i evaluated within coil j

$i, j = 1, 2$

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- Mutual Inductance 互感, M_{12}



In this case, **current in Coil 2 is turned off**, leaving only the flux density generated by Coil 1, $\mathbf{B}_{12}$, existing within Coil 2.

The **mutual linkage** between Coils 1 and 2 is found through:

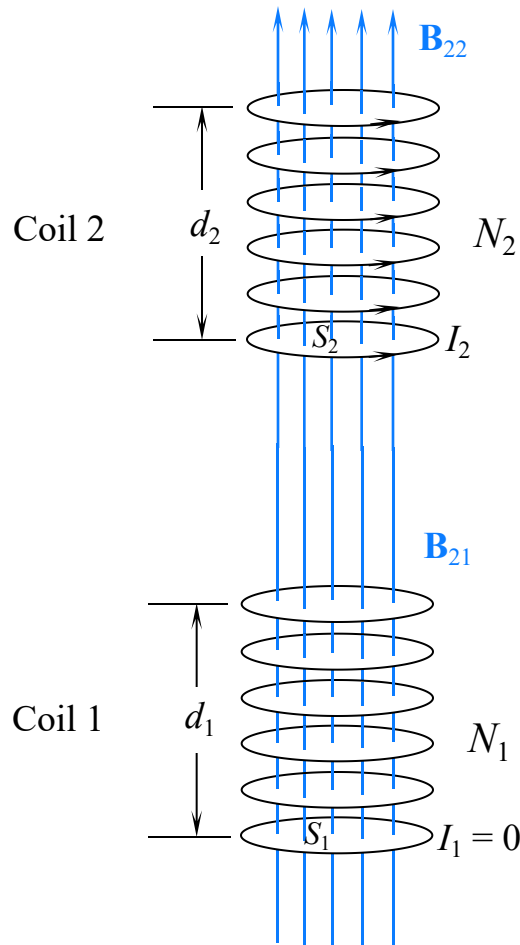
$$\lambda_{12} = N_2 \Phi_{12} = N_2 \int_{S_2} \mathbf{B}_{12} \cdot d\mathbf{S}_2$$

... and **the mutual inductance between Coils 1 and 2 is defined as:**

$$M_{12} \equiv \frac{\lambda_{12}}{I_1}$$

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- Mutual Inductance, M_{21}



In this case, **current in Coil 1 is turned off**, leaving only the flux density generated by Coil 2, $\mathbf{B}_{21}$, existing within Coil 1.

The **mutual linkage** between Coils 2 and 1 is found through:

$$\lambda_{21} = N_1 \Phi_{21} = N_1 \int_{S_1} \mathbf{B}_{21} \cdot d\mathbf{S}_1$$

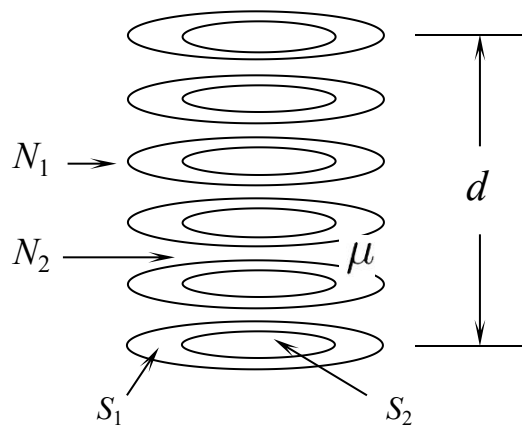
... and the **mutual inductance** between Coils 2 and 1 is defined as:

$$M_{21} \equiv \frac{\lambda_{21}}{I_2}$$

Q: What are the key factors affecting mutual inductance?

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- Example: Concentric Solenoids 同轴螺线管

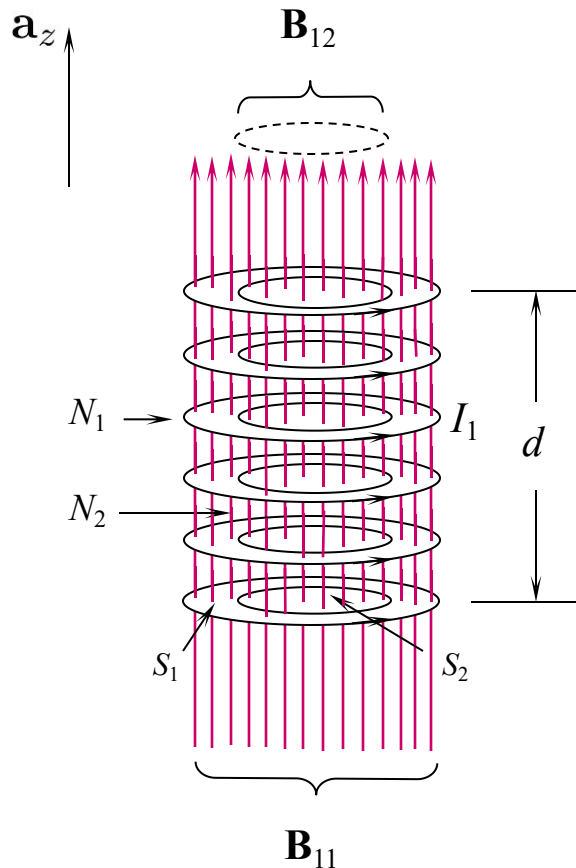


In this configuration, two **concentric solenoids** have different numbers of turns, N_1 and N_2 (even though in the drawing the turn count would appear to be the same). Both coils have the same length, d . The area of each identical turn in the two coils is S_1 for the outer coil, and S_2 for the interior coil. The core permeability is μ



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- Mutual Inductance, M_{12}



With the outer coil current I_1 turned on, the interior flux exists throughout the outer coil volume, and consequently throughout the volume of the inner coil as well.

Flux density $\mathbf{B}_{12}$ resides inside coil 2 (and in coil 1 as well)

Assuming a long coil, $\mathbf{B}_{12} = \frac{\mu N_1 I_1}{d} \mathbf{a}_z$
the flux density is:

The mutual linkage between coils 1 and 2 is then:

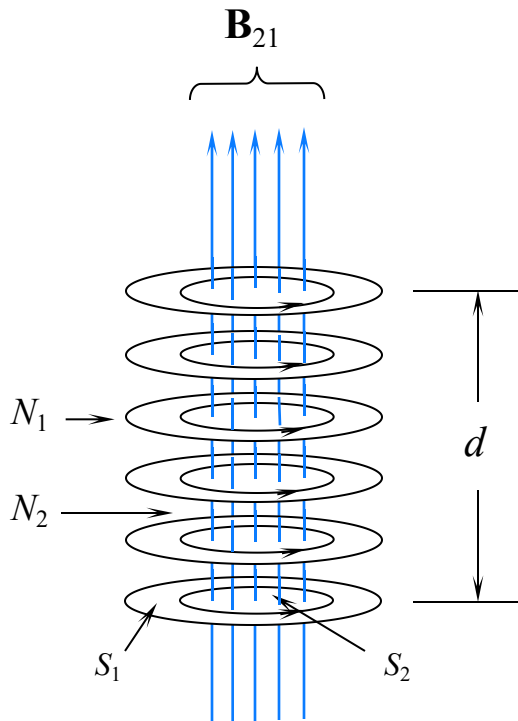
$$\lambda_{12} = N_2 \Phi_{12} = N_2 \frac{\mu N_1 I_1}{d} S_2$$

.. and the mutual inductance is:

$$M_{12} = \frac{\lambda_{12}}{I_1} = N_1 N_2 \frac{\mu S_2}{d}$$

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- Mutual Inductance, M_{21}



With the inner coil current I_2 turned on, the interior flux exists throughout the inner coil volume, which overlaps with the outer coil volume

Flux density $\mathbf{B}_{21}$ resides inside coil 2 (and in coil 1 as well), but the flux in the coil 1 volume is confined within the volume of coil 2.

The coil 2 flux density (that resides in both coils) is:

$$\mathbf{B}_{21} = \frac{\mu N_2 I_2}{d} \mathbf{a}_z$$

The mutual flux linkage between coils 2 and 1 is then:

$$\lambda_{21} = N_1 \Phi_{21} = N_1 \frac{\mu N_2 I_2}{d} S_2$$

...and the mutual inductance is:

$$M_{21} = \frac{\lambda_{21}}{I_2} = N_1 N_2 \frac{\mu S_2}{d} = M_{12}$$

11: The Steady Magnetic Field



- A Property of the Mutual Inductances

The foregoing example illustrates an important property of the mutual inductances between any pair of inductors:

They're equal!

$$M_{12} = M_{21}$$

2. Compare the magnitudes of the electric and magnetic forces on an **electron** that has attained a velocity of 10^7 m/s. Assume an electric field intensity of 10^5 V/m, and a magnetic flux density associated with that of the Earth's magnetic field in temperate latitudes, 0.5 gauss. (Hints: 1. Review previous lecture notes to get the charge of electron; 2. Google the relationship between gauss and tesla as two units for the magnetic flux density.)
3. A point charge for which $Q = 2 \times 10^{-16}$ C and $m = 5 \times 10^{-26}$ kg is moving in the combined fields $\mathbf{E} = 100\mathbf{a}_x - 200\mathbf{a}_y + 300\mathbf{a}_z$ V/m and $\mathbf{B} = -3\mathbf{a}_x + 2\mathbf{a}_y - \mathbf{a}_z$ mT. If the charge velocity at $t = 0$ is $\mathbf{v}(0) = (2\mathbf{a}_x - 3\mathbf{a}_y - 4\mathbf{a}_z)10^5$ m/s (a) give the unit vector showing the direction in which the charge is accelerating at $t = 0$; (b) find the kinetic energy of the charge at $t = 0$.
4. A solenoid is 25 cm long, 3 cm in diameter, and carries 4 A dc in its 400 turns. Its axis is perpendicular to a uniform magnetic field of 0.8 Wb/m² in air. **Using an origin at the center of the solenoid**, calculate the torque acting on it.
5. A toroidal core has a rectangular cross section defined by the surfaces $\rho = 2$ cm, $\rho = 3$ cm, $z = 4$ cm, and $z = 4.5$ cm. The core material has a relative permeability of 80. If the core is wound with a coil containing 8000 turns of wire, find its inductance.